

Week 14 – Capstone Progress Documentation

Project ID: PW26_MG_02

Overview

Week 14 marks a transition from expanding analytical capabilities to strengthening the system's underlying infrastructure. As the project grows in complexity, the limitations of earlier integrations are becoming more apparent. The Zerodha API, while suitable for initial experimentation, is beginning to act as a constraint due to request limits and latency in derivatives data retrieval. This phase feels less like adding new components and more like reinforcing the system's operational backbone—ensuring that data acquisition and processing remain efficient as analytical demands increase.

Progress Update

A key development this week is the integration of the ICICI Breeze API, representing a shift toward a more stable and structured data source. The new integration is improving request handling and reducing latency, enabling smoother execution of repeated queries required for statistical modelling and strategy simulations. This enhancement is allowing the analytical modules to operate with greater consistency.

Under the guidance of our capstone mentor, the system is also evolving toward a multimodal framework. Efforts are underway to integrate diverse data forms, including textual sentiment, numerical time-series, and structured indicators, into a unified analytical pipeline. This reflects a move toward more comprehensive market understanding.

Simultaneously, backend optimizations are being prioritized. Issues related to object hydration in ORM frameworks are being addressed to reduce memory overhead and improve query efficiency. Additionally, exploration of MySQL's InnoDB engine is helping refine indexing strategies and transaction reliability, strengthening the system's persistence layer.

Upcoming Work

The next phase will focus on expanding multimodal integration and refining statistical evaluation frameworks for improved robustness and risk assessment. Backend improvements will continue, particularly in ORM optimization and database efficiency. The system is gradually evolving into a cohesive trading intelligence framework, integrating data, computation, and infrastructure into a unified and efficient research environment.

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